

C-IMMSIM simulation results

January 5, 2026

Abstract

This document includes the plots relative to the simulation and the outcome of the epitope/peptide prediction used.

Produced by the C-IMMSIM Online server available at
<http://c-immsim.iac.rm.cnr.it> (alias to <http://kraken.iac.rm.cnr.it/C-IMMSIM>)

CITATIONS: For publication of results, please cite the following:

Nicolas Rapin, Ole Lund, Massimo Bernaschi, Filippo Castiglione. Computational Immunology Meets Bioinformatics: The Use of Prediction Tools for Molecular Binding in the Simulation of the Immune System. PLoS ONE 5(4): e9862
doi:10.1371/journal.pone.0009862, 2010

A retrospective validation

In-silico evaluation of adenoviral COVID-19 vaccination protocols: Assessment of immunological memory up to 6 months after the third dose. P. Stolfi, F. Castiglione, E. Mastrostefano, I. Di Biase, S. Di Biase, G. Palmieri, A. Prisco. Frontiers in Immunology, 13 (2022) doi: 10.3389/fimmu.2022.998262
<https://www.frontiersin.org/articles/10.3389/fimmu.2022.998262>

An in vivo validation

Identification and validation of viral antigens sharing sequence and structural homology with tumor associated antigens (TAAs). C. Ragone, C. Manolio, B. Cavalluzzo, A. Petrizzo, A. Mauriello, M-L. Tornesello, F. M. Buonaguro, F. Castiglione, L. Vitagliano, M. Ruvo, M. Tagliamonte, L. Buonaguro. Journal for ImmunoTherapy of Cancer. 9:e002694 (2021)
<https://jitc.bmj.com/content/9/5/e002694>

Original C-IMMSIM model: www.iac.cnr.it/~filippo/c-immsim

GETTING HELP:

Scientific problems: Filippo Castiglione (filippo dot castiglione at cnr dot it)

Technical problems: Ilaria Gonnella (ilaria dot gonnella at cnr dot it)

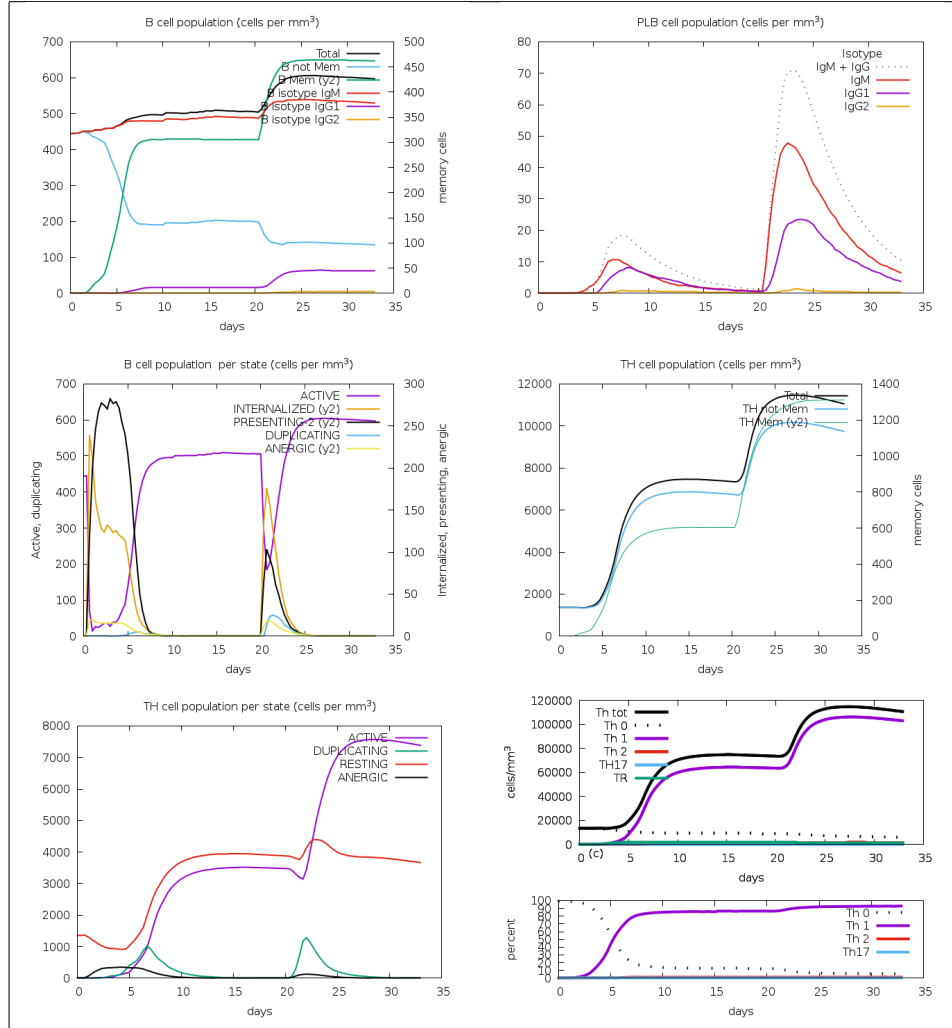


Figure 1: Cell counts shown. Legend: Act=active, Intern=internalized the Ag, Pres II = presenting on MHC II, Dup = in the mitotic cycle, Anergic = anergic, Resting = not active.

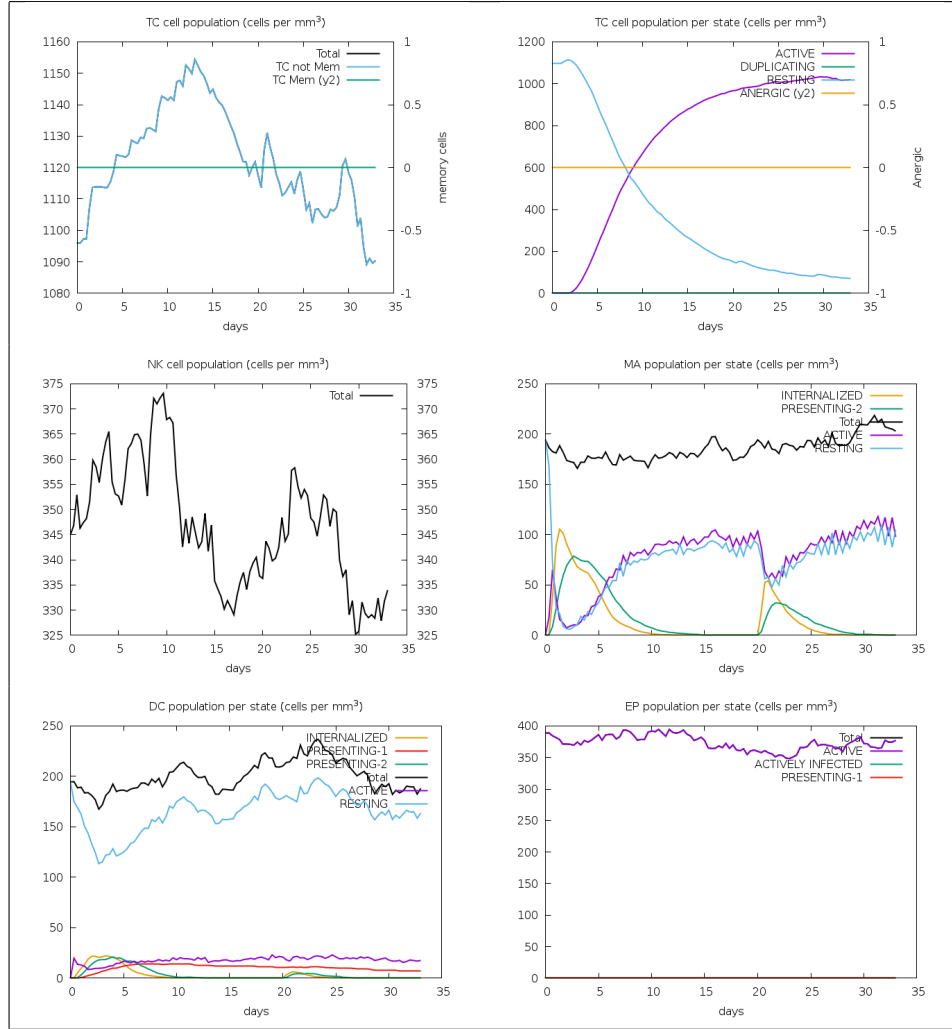


Figure 2: Legend: symbols as figure above.

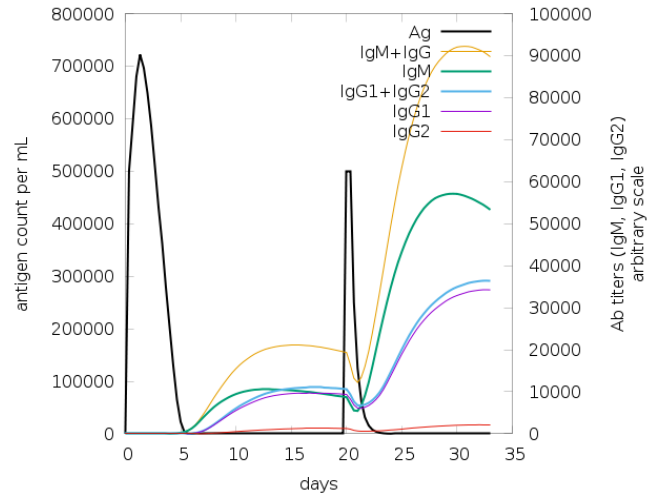


Figure 3: The virus, the immunoglobulins and the immunocomplexes.

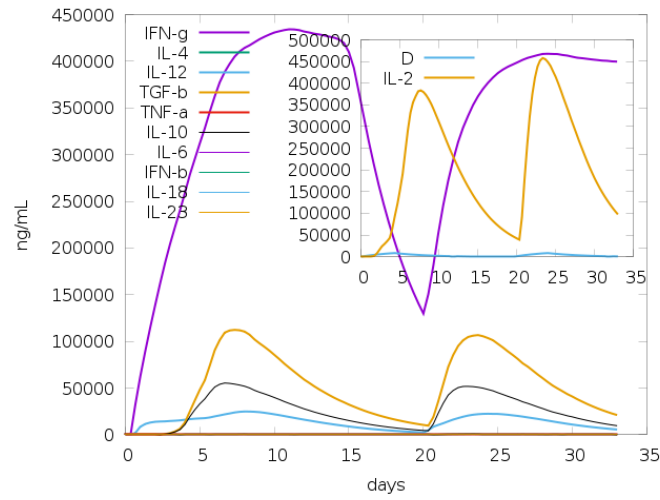


Figure 4: Concentration of cytokines and interleukins. Inset plot shows danger signal together with leukocyte growth factor IL-2.

Use Parker's propensity scale, takes an antigen block as input, and creates a list of residues that are possible epitopes.

AKFVAAWTLKAAAGGSCPSLDVDSIIPRTPDVLHEDAAAAGPGTGPNGLGEKGDTSGPEGSGGGSPQRRGGDNHGRGR
GRGRGRGGGRPGAPGGSGS GPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGG
AGAGGGAGAGGGAGGAGGAGAGGGAGAEEAAAKRVLSSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPELKGPGPLPQQQ
LTAYGPGPGFLDKGYTLGPGPGLEKQLFYIIGTMLPNTRPHEAAAKPPMVEGAAGGDDGDGPGPGSDREVRGS

[illegible]

```

1]      pos=41 len=6      GPGTGP
2]      pos=50 len=53     LGEKGDTSGPEGSGSGPQRRGGDNHGRGRGRGRGRGGRRPGAPGSGSGSPRH
3]      pos=104 len=4     DGVR
4]      pos=120 len=66     GTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGGAGAGGGAGAGGGAGGAGAGGGAG
5]      pos=295 len=20     AAEGDDGDGPGPGSDREVRG

```

Given the antigen injected creates the list of peptides for all the NumAgProts proteins and for all i.e., 4 MHC I molecules

Read class I peptide list from file? NO

Allele: A0101
Pseudo sequence: KAVHAEQRNKAQTRA
Threshold: 9.456400
Max score: 29.236000

AKFVAAWTLKAAAGGSCPSLDVDSIIPRTPDVLHEDEAAAKGPGTGPNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGR.
GRGRGRGGGRPGAPGGSGSGPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGGAGGGAGAGGGAGGAGGAGG
AGAGGGAGAGGGAGGAGGAGAGGGAGAGAAAKRVLSSVVLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGLPQQQ
LTAYGPGPGFLDKGTYTLGPGPGPLEKQLFYIIGTMLPNTRPHEAAAKPPMVEGAAAGDDGDGPGPGSDREVRGS

```
0] pos= 263 score=0.033756 unnormalised=3.6626000000 PLEKQLFYF
1] pos= -1 score=0.966244 unnormalised=104.8410000000 non-binding event
```

5

Max score: 29.236000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-012817_5_KDx1WNC7F5NM.FSA_1_001

AKFVAAWTLKAAAGGSCPSLDVDSIIPRTDVLHEDEAAAKGPGTGPNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGR
GRGRGRGGGRPGAPGGSGSPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGAGAGGGAGAGGGAGGAGGAGG
AGAGGGAGAGGGAGGAGAGGGAGAGAAAKRVLSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGLPQGQ
LTAYGPGPGFLDKGTYTLGPGPGPLEKQLFYYIGTMLPNTRPHEAAAKPPMVEGAAAEAGDDGDGPGPGSDREVRGS

Epitopes of protein 0 -----

0]	pos= 263	score=0.033756	unnormalised=3.6626000000	PLEKQLFYY
1]	pos= -1	score=0.966244	unnormalised=104.8410000000	non-binding event

=====

Allele: B0702
Pseudo sequence: KAAREEQIKAQTRE
Threshold: 8.702800
Max score: 28.406000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-012817_5_KDx1WNC7F5NM.FSA_1_001

AKFVAAWTLKAAAGGSCPSLDVDSIIPRTDVLHEDEAAAKGPGTGPNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGR
GRGRGRGGGRPGAPGGSGSPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGAGAGGGAGAGGGAGGAGGAGG
AGAGGGAGAGGGAGGAGAGGGAGAGAAAKRVLSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGLPQGQ
LTAYGPGPGFLDKGTYTLGPGPGPLEKQLFYYIGTMLPNTRPHEAAAKPPMVEGAAAEAGDDGDGPGPGSDREVRGS

Epitopes of protein 0 -----

0]	pos= 89	score=0.002703	unnormalised=0.3012000000	RPGAPGGSG
1]	pos= 108	score=0.017339	unnormalised=1.9322000000	RPQKRPSCI
2]	pos= 232	score=0.014064	unnormalised=1.5672000000	GPGLPQGQL
3]	pos= 276	score=0.025083	unnormalised=2.7952000000	LPNTRPHEA
4]	pos= -1	score=0.940811	unnormalised=104.8410000000	non-binding event

=====

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AKFVAAWTLKAAAGGSCPSLDVDSIIPRTDVLHEDEAAAKGPGTGPNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGR
GRGRGRGGGRPGAPGGSGSPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGAGAGGGAGAGGGAGGAGGAGG
AGAGGGAGAGGGAGGAGAGGGAGAGAAAKRVLSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGLPQGQ
LTAYGPGPGFLDKGTYTLGPGPGPLEKQLFYYIGTMLPNTRPHEAAAKPPMVEGAAAEAGDDGDGPGPGSDREVRGS

Epitopes of protein 0 -----

0]	pos= 89	score=0.002703	unnormalised=0.3012000000	RPGAPGGSG
1]	pos= 108	score=0.017339	unnormalised=1.9322000000	RPQKRPSCI
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3]	pos= 276	score=0.025083	unnormalised=2.7952000000	LPNTRPHEA
4]	pos= -1	score=0.940811	unnormalised=104.8410000000	non-binding event

DoPeptideList_II:

Given the antigen injected creates the list of peptides for all the
NumAgProts proteins and for all i.e., 2 MHCII molecules

Read class II peptide list from file? NO

Allele: DRB1_0101
Pseudo sequence: KAFHVEQRKAQTRV
Threshold: 2.392440
Max score: 26.461000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-012817_5_KDx1WNC7F5NM.FSA_1_001

AKFVAAWTLKAAAGGSCPSLDVDSIIIPRTDVLHEDEAAAKGPGTGPNGLGEKGDTS GPEGSGGSGPQRRGGDNHGRGR
GRGRGRGGGRPGAPGGSGSGPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGAGGGAGAGGGAGGAGGAGG
AGAGGGAGAGGGAGGAGGAGAGGAGAGGAGAAAKRVLSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGLPQGQ
LTAYGPGPGLDKGTYTLGPGPGPLEKQLFFYYIGTMLPNTRPHEAAAKPPMVEGAAAEAGDDGDGPGPGSDREVGRS

Epitopes of protein 0 -----

0]	pos=	2	score=0.041679	unnormalised=3.3845600000	FVAAWTLKA
1]	pos=	6	score=0.005684	unnormalised=0.4615600000	WTLKAAAGG
2]	pos=	197	score=0.020609	unnormalised=1.6735600000	VVLLGPGPG
3]	pos=	198	score=0.010487	unnormalised=0.8515600000	VLLGPGPGT
4]	pos=	214	score=0.036076	unnormalised=2.9295600000	FGPGPGACM
5]	pos=	227	score=0.010979	unnormalised=0.8915600000	LEKGPGL
6]	pos=	269	score=0.034894	unnormalised=2.8335600000	FFYYIGTMLP
7]	pos=	272	score=0.024759	unnormalised=2.0105600000	IGTMLPNTR
8]	pos=	-1	score=0.814832	unnormalised=66.1680000000	non-binding event

Allele: DRB1_0101
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Threshold: 2.392440
Max score: 26.461000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-012817_5_KDx1WNC7F5NM.FSA_1_001

AKFVAAWTLKAAAGGSCPSLDVDSIIIPRTDVLHEDEAAAKGPGTGPNGLGEKGDTS GPEGSGGSGPQRRGGDNHGRGR
GRGRGRGGGRPGAPGGSGSGPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGAGGGAGAGGGAGGAGGAGG
AGAGGGAGAGGGAGGAGGAGGAGAGGAGAAAKRVLSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGLPQGQ
LTAYGPGPGLDKGTYTLGPGPGPLEKQLFFYYIGTMLPNTRPHEAAAKPPMVEGAAAEAGDDGDGPGPGSDREVGRS

Epitopes of protein 0 -----

0]	pos=	2	score=0.041679	unnormalised=3.3845600000	FVAAWTLKA
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3]	pos=	198	score=0.010487	unnormalised=0.8515600000	VLLGPGPGT
4]	pos=	214	score=0.036076	unnormalised=2.9295600000	FGPGPGACM
5]	pos=	227	score=0.010979	unnormalised=0.8915600000	LEKGPGL
6]	pos=	269	score=0.034894	unnormalised=2.8335600000	FFYYIGTMLP
7]	pos=	272	score=0.024759	unnormalised=2.0105600000	IGTMLPNTR
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